

# Completed $4N_y$ hard-wall dynamical-critical analysis

Numerical working note  
(Dated: September 17, 2026)

We analyze 700 independent Born trajectories of the  $N_x = 20$  hard-wall adaptive Gaussian circuit at  $N_y = 20, 24, 30, 36, 44, 56, 60$  and depth  $T = 4N_y$ . All 140 five-sample shards pass checksum, identity, coverage, and numerical validation. The saved occupation spectra reproduce the stored leading many-body squared-singular-value levels to a maximum absolute error of  $7.28e - 12$ . The calculation robustly establishes purification, record-weight closure, and boundary localization. The preregistered finite-size gates are applied without forcing a central-charge or operator-spectrum claim.

## I. PROTOCOL AND RECONSTRUCTION

The campaign uses maximally mixed initial conditions, hard support-truncated walls at  $x = 5, 15$ ,  $n_{\text{shell}} = 1$ ,  $\alpha_1 = 1$ ,  $\alpha_2 = 30$ , raster- $y$  measurements, perfect correction, and complex128 covariance evolution. Entropies and active-space occupations were saved every four cycles; the realized Born record weight  $\omega(t)$  was saved every cycle. For each saved occupation spectrum  $\{\nu_a\}$  we reconstruct the many-body levels

$$\ell_i = \log \sigma_i^2 = \log Z + \sum_a \log \max(\nu_a, 1 - \nu_a) - d_i, \quad (1)$$

where  $d_i$  are the lowest subset sums of  $|\log[\nu_a/(1 - \nu_a)]|$ . Exact zero and unit occupations retain infinite flip costs. We fit every trajectory separately in  $W_1 = [N_y, 2N_y]$ ,  $W_2 = [2N_y, 3N_y]$ , and  $W_3 = [3N_y, 4N_y]$ . All confidence intervals resample complete trajectories, with 2000 replicates and seed 2026091702.

## II. PURIFICATION AND TEMPORAL CONVERGENCE

The endpoint entropy is small at every circumference, while the full  $S(t)/N_y$  curves provide a direct descriptive test of  $z = 1$  scaling. Threshold-crossing times are linearly interpolated on the native four-cycle grid; an exploratory exponent is withheld unless the three thresholds and omitted-smallest-size fit agree. The combined late temporal gate is **failed**. This gate requires the paired  $W_2 \rightarrow W_3$  confidence interval to contain zero, a mean shift below 10%, and at least 95% resolved trajectories for every reported metric.

## III. RECORD CLOSURE AND BOUNDARY LOCALIZATION

The leading reconstructed level is compared with  $d\omega/dt$  in the same fit window. The endpoint ratio  $-\omega(T)/T$  is retained only as a finite-depth diagnostic. The  $W_3$  record-closure ledger **passed** its one-percent criterion across all sizes. The first four soft modes are

analyzed only through their aggregate subspace density and aggregate weight near both walls. This avoids nonphysical labels under rotations in nearly degenerate eigenspaces.

## IV. FINITE-SIZE COEFFICIENTS AND CLAIM GATES

The late-window leading density is fit to

$$-\lambda_0/N_y = f_\infty + A_0/N_y^2, \quad (2)$$

with  $1/N_y^4$  and successive  $N_{y,\text{min}} = 24, 30, 36$  sensitivity fits. The primary result is  $A_0 = 11.19$  with 95% interval  $[-4.148, 27.04]$ . The derived  $\alpha c_{\text{eff}} = -6A_0/\pi$  is **not released**; release requires the physical sign, bootstrap significance, temporal stability, and below-20% fit sensitivity simultaneously. The same rules, plus resolved-sample gates, apply to  $\alpha x_i = A_i/(2\pi)$ . Released ordered gaps: none. No conformal operator names are assigned because momentum, charge-sector, and two-wall symmetry labels were not saved.

## V. RELATION TO EARLIER ENSEMBLES

The earlier  $2N_y$  pilot and the independent three-size  $4N_y$  purification campaign are compared only at the summary-ledger level. Their trajectories are never pooled with this primary seven-size campaign.

## VI. CONCLUSION

The completed data validate the Gaussian many-body spectral reconstruction, deep-time purification, same-window record closure, and boundary localization. Any failed gate remains explicit in the accompanying CSV and JSON ledgers; an unphysical or unstable coefficient is not converted into a claim about  $c_{\text{eff}}$  or absolute operator dimensions.

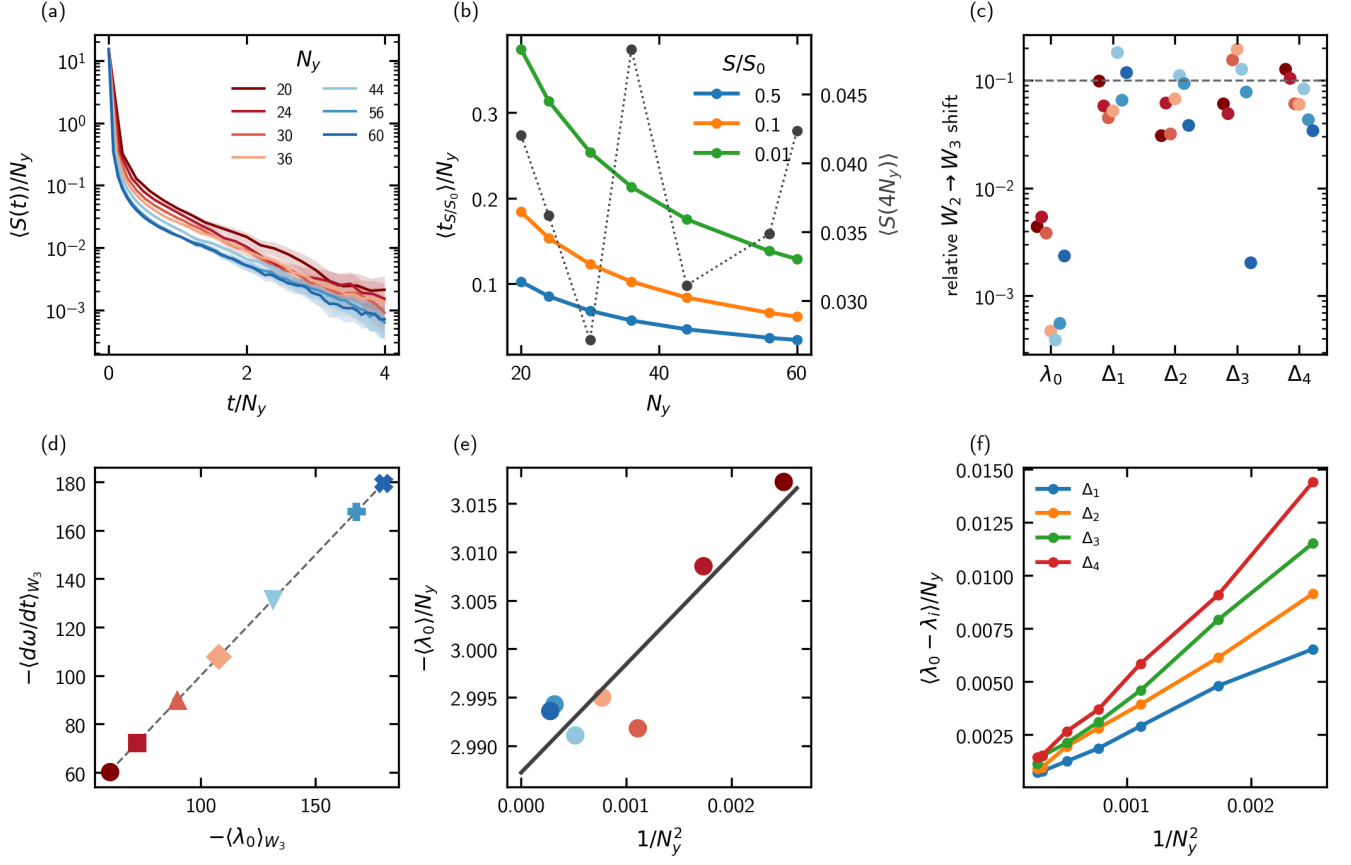


FIG. 1. Primary dynamical-critical results. (a) Entropy-density collapse with whole-trajectory bootstrap intervals. (b) Sample-wise purification times and endpoint residual entropy. (c) Relative  $W_2 \rightarrow W_3$  shifts. (d) Same-window record-weight closure. (e) Leading finite-size fit. (f) Ordered-gap densities. Colors label circumference.

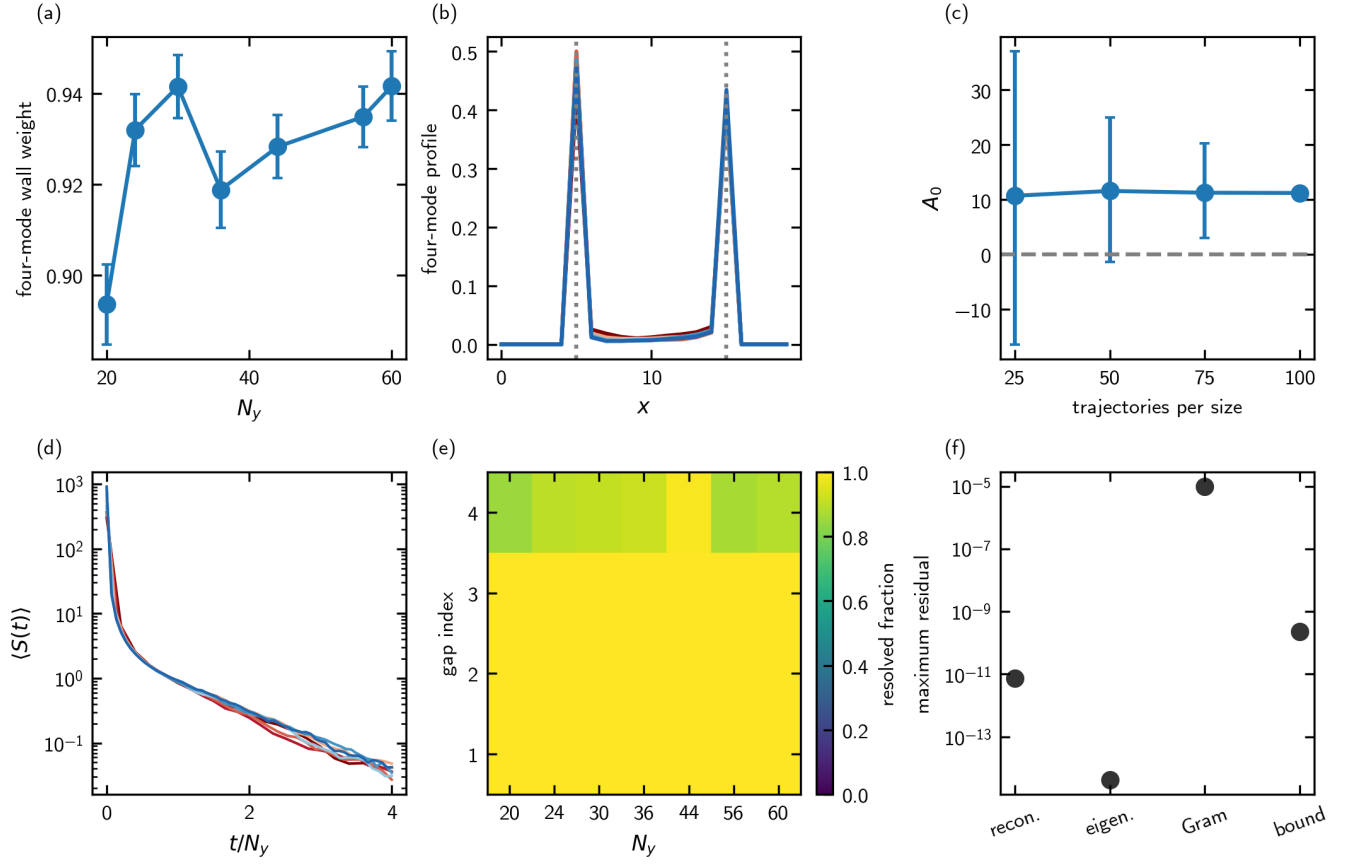


FIG. 2. Diagnostics. (a,b) Aggregate four-mode wall localization and transverse profile. (c) Random without-replacement sample convergence. (d) Unnormalized entropy dynamics. (e) Resolved fractions. (f) Maximum numerical residuals.